

We now wish to compute the output noise spectrum of a sampling mixer. The output noise at the IF can then be divided by the conversion gain to obtain the input-referred noise voltage. We begin with three observations. First, in the simple circuit of Fig. 6.29(a) (where R_1 denotes the switch resistance), if $V_{in} = 0$,

$$\overline{V_{n,LPF}^2} = \overline{V_{nR1}^2} \frac{1}{1 + (R_1 C_1 \omega)^2}, \quad (6.30)$$

where $\overline{V_{nR1}^2} = 2kTR_1$ (for $-\infty < \omega < +\infty$). We say the noise is “shaped” by the filter.³ Second, in the switching circuit of Fig. 6.29(b), the output is equal to the shaped noise of

R_1 when S_1 is on and a *sampled*, constant value when it is off. Third, in a manner similar to the gain calculation in Fig. 6.21, we can decompose the output into two waveforms V_{n1} and V_{n2} as shown in Fig. 6.29(c).

It is tempting to consider the overall output spectrum as the sum of the spectra of V_{n1} and V_{n2} . However, as explained below, the low-frequency noise components generated by R_1 create *correlation* between the track-mode and hold-mode noise waveforms. For this reason, we proceed as follows: (1) compute the spectrum of V_{n1} while excluding the low-frequency components in the noise of R_1 , (2) do the same for V_{n2} , and (3) add the contribution of the low-frequency components to the final result. In the derivations below, we refer to the first two as simply the spectra of V_{n1} and V_{n2} even though $V_{n1}(t)$ and $V_{n2}(t)$ in Fig. 6.29 are affected by the low-frequency noise of R_1 . Similarly, we use the notation $\overline{V_{n,LPF}^2}(f)$ even though its low-frequency components are removed and considered separately.

Spectrum of V_{n1} To calculate the spectrum of V_{n1} , we view this waveform as the product of $V_{n,LPF}(t)$ and a square wave toggling between 0 and 1. As shown in Fig. 6.30, the spectrum of V_{n1} is given by the convolution of $\overline{V_{n,LPF}^2}(f)$ and the power spectral density of the square wave (impulses with a sinc^2 envelope). In practice, the sampling bandwidth of the mixer, $1/(R_1 C_1)$, rarely exceeds $3\omega_{LO}$, and hence

$$\overline{V_{n1}^2}(f) = 2 \times \left(\frac{1}{\pi^2} + \frac{1}{9\pi^2} \right) \frac{2kTR_1}{1 + (2\pi R_1 C_1 f)^2}, \quad (6.31)$$

where the factor of 2 on the right-hand side accounts for the aliasing of components at negative and positive frequencies. At low output frequencies, this expression reduces to

$$\overline{V_{n1}^2} = 0.226(2kTR_1). \quad (6.32)$$

Note that this is the two-sided spectrum of $\overline{V_{n1}^2}$.

- 6.4. We express V_{n1} in Fig. 6.29(c) as the product of the shaped resistor noise *voltage* and a square wave toggling between 0 and 1. Prove that the spectrum of V_{n1} is given by Eq. (6.31).

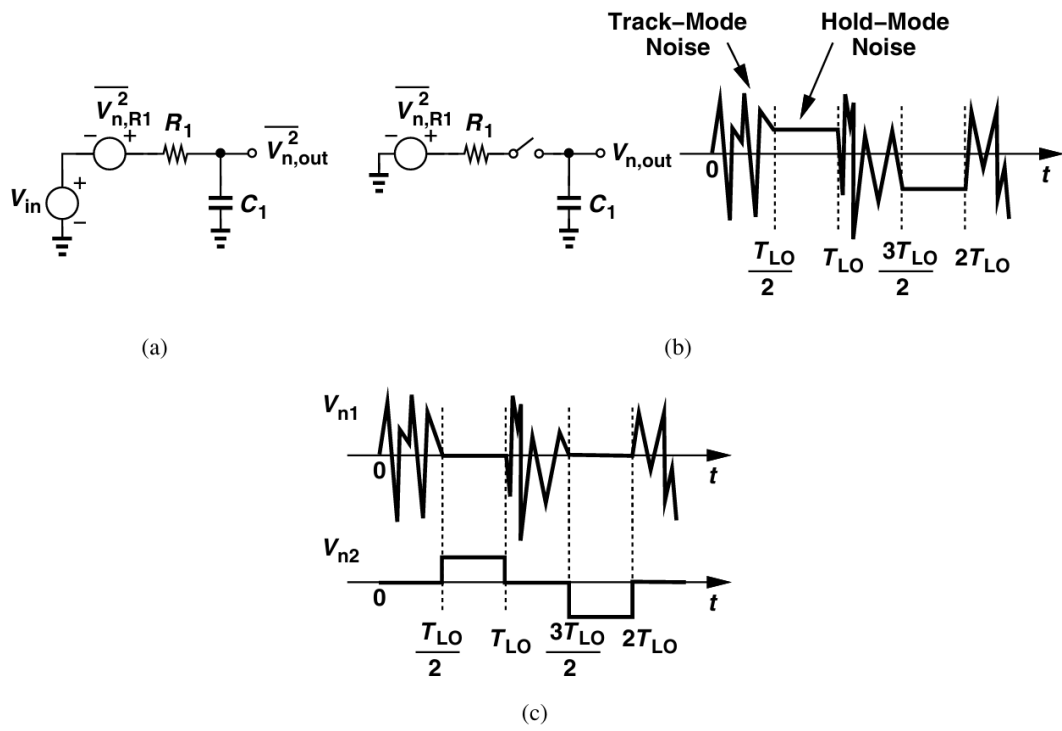


Figure 6.29 (a) Equivalent circuit of sampling mixer for noise calculations, (b) noise in on and off states, and (c) decomposition of output waveform.

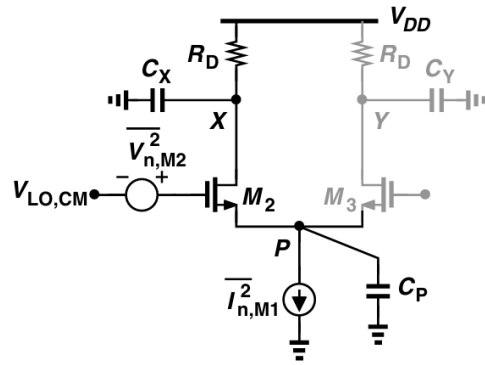


Figure 6.57 Noise of input device and one switching device in an active mixer.

- 6.7. In the active mixer of Fig. 6.57, $I_{n,M1}$ contains all frequency components. Prove that the convolution of these components with the harmonics of the LO in essence multiplies $4kT\gamma/g_m$ by a factor of $\pi^2/4$.

- 6.9. Shown in Fig. 6.100 is the front end of a 1.8-GHz receiver. The LO frequency is chosen to be 900 MHz and the load inductors and capacitances resonate with a quality factor of Q at the IF. Assume M_1 is biased at a current of I_1 , and the mixer and the LO are perfectly symmetric.
- (a) Assuming M_2 and M_3 switch abruptly and completely, compute the LO-IF feedthrough, i.e., the measured level of the 900-MHz output component in the absence of an RF signal.
- (b) Explain why the flicker noise of M_1 is critical here.

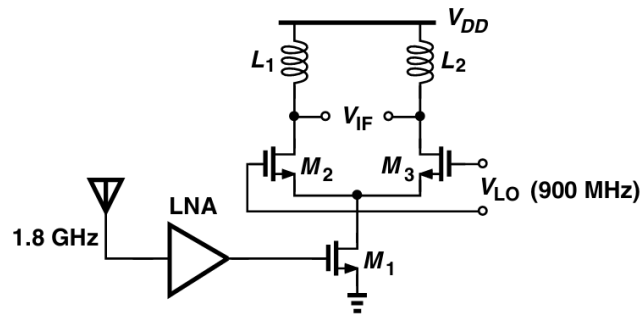


Figure 6.100 Front-end chain for a 1.8-GHz RX.

- 6.16. The circuit shown in Fig. 6.101 is a dual-gate mixer used in traditional microwave design. Assume when M_1 is on, it has an on-resistance of R_{on1} . Also, assume abrupt edges and a 50% duty cycle for the LO and neglect channel-length modulation and body effect.

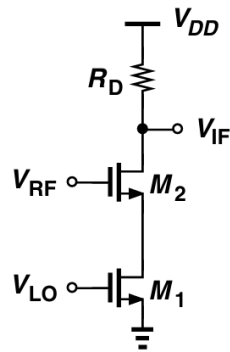


Figure 6.101 *Dual-gate mixer.*

- Compute the voltage conversion gain of the circuit. Assume M_2 does not enter the triode region and denote its transconductance by g_{m2} .
- If R_{on1} is very small, determine the IP_2 of the circuit. Assume M_2 has an overdrive of $V_{GS0} - V_{TH}$ in the absence of signals (when it is on).

- 6.17. Consider the active mixer shown in Fig. 6.102, where the LO has abrupt edges and a 50% duty cycle. Also, channel-length modulation and body effect are negligible. The load resistors exhibit mismatch, but the circuit is otherwise symmetric. Assume M_1 carries a bias current of I_{SS} .
- (a) Determine the output offset voltage.
- (b) Determine the IP_2 of the circuit in terms of the overdrive and bias current of M_1 .

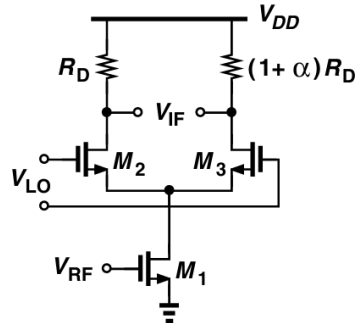


Figure 6.102 Active mixer with load mismatch.